

TikTok Claims Classification Project

Executive Summary — Course 1: Foundations of Data Science

Proprietary + Confidential

Overview

This project aims to support TikTok's Trust & Safety team by building a machine learning model that classifies whether statements made in videos are claims or opinions. Course 1 focuses on the initial data inspection and organization phase — understanding the dataset before any modeling begins.

The Problem

TikTok receives large volumes of user-generated videos containing unverified claims. Manually reviewing these at scale is not feasible. The data team needs to:

- Understand the volume and distribution of claims vs. opinions in the dataset
- Identify which variables correlate with claim status and engagement
- Prepare the data for more advanced EDA, hypothesis testing, and eventual ML modeling

The Solution

A data professional was assigned to inspect, organize, and summarize TikTok's provided dataset using Python (pandas and numpy). The workflow followed the PACE framework: Plan → Analyze → Construct → Execute.

Details

Task	Method	Output
Build dataframe	pd.read_csv() to load tiktok_dataset.csv	19,383-row DataFrame with 12 columns
Inspect data	data.head(10), data.info(), data.describe()	Data types, null counts, and statistical summaries identified
Examine variables	value_counts(), groupby(), min/max analysis	Claim/opinion split, engagement patterns, null distribution confirmed
Engineer features	Created likes_per_view, comments_per_view, shares_per_view	Normalized engagement rates enabling fair comparison across videos

Keys to success

- Near-balanced target variable: ~49.6% claims vs ~50.4% opinions — favorable for unbiased ML classification
- Null values are systematic (all 5 engagement columns missing together for the same 298 rows) — likely a data collection issue, not random
- Claim status drives engagement rate more than raw volume — a critical distinction for feature selection
- Transcription text will be essential for NLP-based classification and should be prioritized in EDA

Data variable assessment

- video_duration_sec: min = 5s, max = 60s — no anomalies
- video_view_count: min ≈ 20, max ≈ 999,817 — heavy right skew
- Relevant columns: claim_status (target), video_transcription_text (NLP), author_ban_status, all engagement metrics
- Less relevant: # (index), video_id (identifier only)

Results Summary

The dataset was successfully loaded, inspected, and organized. The near-balanced claim/opinion split (49.6% vs 50.4%) is favorable for machine learning. Engagement analysis reveals that claim videos consistently generate higher engagement rates than opinion videos — a pattern that holds across all author ban status categories and will likely be a strong predictive signal.

Finding	Detail
Dataset: 49.6% claims, 50.4% opinions	Near-balanced — favorable for ML classification
298 rows with null engagement values	Systematic nulls — investigate before modeling
Claim videos: likes_per_view ≈ 0.33	vs opinion videos ≈ 0.22 — 50% higher engagement rate
Banned authors: ~2x more raw views	But claim status is stronger driver of engagement rate
video_transcription_text identified as key	Central to planned NLP classification approach

Reflections / Next Steps

- Handle the 298 null rows — confirm whether systematic or random, then drop or impute before proceeding to EDA
- Conduct full EDA with visualizations of engagement distributions segmented by claim_status and author_ban_status
- Prioritize NLP exploration of video_transcription_text — text features will likely be the most direct signal for classification
- Quantify the relationship between author_ban_status and claim_status — these appear correlated and may introduce multicollinearity in modeling
- Begin feature engineering pipeline in preparation for hypothesis testing and regression analysis in Course 2